

Rohit Goswami

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WORK EXPERIENCE

LAB COSMO, EPFL | SYSTEMS SPECIALIST (POSTDOCTORAL RESEARCHER)

2025 – Present | Lausanne, Switzerland

- Developing the metatensor ecosystem for interoperable atomistic machine learning with Prof. Michele Ceriotti and Dr. Guillaume Fraux
- Integration of machine learning potentials with GROMACS; DLPack support for zero-copy tensor interoperability across C++, Python, Rust, and Fortran

QUEEN'S UNIVERSITY | VISITING RESEARCHER

2025 | Kingston, Canada

- Scaling off-lattice / object kinetic Monte Carlo to over a million atoms for radiation damage modeling with Prof. Laurent B  land

UNIVERSITY OF ICELAND | DOCTORAL RESEARCHER (RANN  S DOCTORAL GRANT PI)

2019 – 2024 | Reykjav  k, Iceland

- Principal investigator: “Magnetic interactions of itinerant electrons modeled using Bayesian machine learning” (Rann  s Research Fund)
- Developed Gaussian process accelerated saddle point search methods (climbing image NEB, dimer) with adaptive pruning
- Bayesian hierarchical models for quantitative benchmarking of saddle search algorithms

QUANSIGHT LABS | SOFTWARE ENGINEER II

2021 – 2025 | Remote (Austin, TX)

- Maintained f2py (NumPy's Fortran-to-Python interface) and foundational FOSS scientific codebases
- Contributed to scipy, meson-python, and LFortran compiler development

IIT KANPUR | SENIOR PROJECT ASSOCIATE / PROJECT ASSOCIATE

2018 – 2020 | Kanpur, India

- Computational nanoscience group: ice nucleation analysis (CHILL+), GPU-accelerated MD tools; created d-SEAMS
- Femtolab: femtosecond laser approaches to quantum information and computation

EDUCATION

UNIVERSITY OF ICELAND | PH.D. PHYSICAL SCIENCES

2019 – 2025 | Reykjav  k, Iceland

Thesis: “Efficient Exploration of Chemical Kinetics”. Supervised by Prof. Hannes J  nsson.

HBTU KANPUR | B.TECH. CHEMICAL ENGINEERING

2014 – 2018 | Kanpur, India

First Division.

SELECTED PUBLICATIONS

- [9] Bigi, Filippo, Abbott, Joseph W., Loche, Philip, Mazitov, Arslan, Tisi, Davide, Langer, Marcel F., Goscinski, Alexander, Pegolo, Paolo, Chong, Sanggyu, **Goswami, Rohit**, Febrer, Pol, Chorna, Sofia, Kellner, Matthias, Ceriotti, Michele and Fraux, Guillaume. **Feb. 2026**. ‘Metatensor and Metatomic: Foundational Libraries for Interoperable Atomistic Machine Learning’. In: *The Journal of Chemical Physics* 164.6. DOI: 10.1063/5.0304911. arXiv: 2508.15704.[PDF]
- [8] **Goswami, Rohit**. **Mar. 2026**. ‘Two-Dimensional RMSD Projections for Reaction Path Visualization and Validation’. In: *MethodsX*, p. 103851. DOI: 10.1016/j.mex.2026.103851. arXiv: 2512.07329.[PDF]
- [7] **Goswami, Rohit**. **11th Aug. 2025**. ‘Bayesian Hierarchical Models for Quantitative Estimates for Performance Metrics Applied to Saddle Search Algorithms’. In: *AIP Advances* 15.8, p. 85210. DOI: 10.1063/5.0283639. arXiv: 2505.13621.[PDF]
- [6] **Goswami, Rohit** and J  nsson, Hannes. **27th Nov. 2025**. ‘Adaptive Pruning for Increased Robustness and Reduced Computational Overhead in Gaussian Process Accelerated Saddle Point Searches’. In: *ChemPhysChem (Cover feature)*. DOI: 10.1002/cphc.202500730. arXiv: 2510.06030.[PDF]
- [5] **Goswami, Rohit**, Masterov, Maxim, Kamath, Satish, Pena-Torres, Alejandro and J  nsson, Hannes. **July 2025**. ‘Efficient Implementation of Gaussian Process Regression Accelerated Saddle Point Searches with Application to Molecular Reactions’. In: *Journal of Chemical Theory and Computation*. DOI: 10.1021/acs.jctc.5c00866. arXiv: 2505.12519.[PDF]
- [4] **Goswami, Rohit**, Rawat, Ashwini Kumar, Goswami, Sonaly and Goswami, Debabrata. **June 2025**. ‘Compositional Analysis of Fragrance Accords Using Femtosecond Thermal Lens Spectroscopy’. In: *Chemistry – an Asian Journal* 20.17. DOI: 10.1002/asia.202500521. arXiv: 2503.20431.[PDF]

- [3] Sallermann, Moritz, Goswami, Amrita, Peña-Torres, Alejandro and **Goswami, Rohit**. Oct. 2025. 'Flowy: High Performance Probabilistic Lava Emplacement Prediction'. In: *Computer Physics Communications* 315, p. 109745. DOI: 10.1016/j.cpc.2025.109745. arXiv: 2405.20144.[PDF]
- [2] Kedward, Laurence, Aradi, Balint, Certik, Ondrej, Curcic, Milan, Ehlert, Sebastian, Engel, Philipp, **Goswami, Rohit**, Hirsch, Michael, Lozada-Blanco, Asdrubal, Magnin, Vincent, Markus, Arjen, Pagone, Emanuele, Pribec, Ivan, Richardson, Brad, Snyder, Harris, Urban, John and Vandenplas, Jeremie. Apr. 2022. 'The State of Fortran'. In: *Computing in Science & Engineering*. DOI: 10.1109/MCSE.2022.3159862. arXiv: 2203.15110.[PDF]
- [1] **Goswami, Rohit**, Goswami, Amrita and Singh, Jayant Kumar. Mar. 2020. 'd-SEAMS: Deferred Structural Elucidation Analysis for Molecular Simulations'. In: *Journal of Chemical Information and Modeling*. DOI: 10.1021/acs.jcim.0c00031. arXiv: 1909.09830.[PDF]

GRANTS AND AWARDS

COST ACTION CA23158 CO-PROPOSER | 575K EUR, 2025–2029

European Cooperation in Science and Technology network proposal.

RANNÍS DOCTORAL RESEARCH GRANT | 6650K ISK, 2020–2023

Icelandic Research Fund grant as PI for Bayesian ML applied to saddle point searches.

NUMFOCUS SMALL DEVELOPMENT GRANT | \$5000 USD, 2020

For d-SEAMS molecular simulation analysis engine.

SPRINGER BEST STUDENT PAPER AWARD | PHOTONICS 2016

SELECTED SERVICE

- **JOSS Editor** (2024–present) – Journal of Open Source Software, topic editor
- **NumFOCUS GSoC Admin** (2024–present) – Coordinating Google Summer of Code across NumFOCUS projects
- **IEEE P3173 Vice Chair** (2022–present) – Standard for Endocrine Disrupting Chemical Hazard Labelling
- **Nordic-RSE** – Board Member; founding Secretary of the Nordic Research Software Engineers association
- **IHPC Consultant** (2021–2024) – Simulation and Data Lab Computational Chemistry, National HPC center of Iceland
- **Reviewer:** Nature Comms, PRL, PRA, PRE, JOSS (80+), PLOS ONE, IOP journals, PeerJ

SELECTED PRESENTATIONS

- **ACS Fall 2025** – “Managing concurrency for high throughput adaptive kinetic Monte Carlo” (Washington, DC)
- **Psi-k 2025** – “Advocating for standardized APIs for common procedures” (Liverpool, UK)
- **Kernel aKMC 2025** – “Towards a million atoms and a thousand KMC steps a day” (Queen’s, Canada)
- **APS Global Physics Summit 2025** – “Contending with concurrency across programming languages” (Anaheim, CA)
- **CECAM-LTS-MAP 2024** – “Accelerating Saddle Searches with Gaussian Process Regression” (Invited, Lausanne)
- **IPAM TMRC2 2023** – “Computational Chemistry and Possible Tensor Applications” (Invited, UCLA)
- **14th Icelandic HPC Workshop 2024** – “Scaling workflows from development to HPC”

OPEN SOURCE SOFTWARE

- **HiGHS / SciPy** (contributor) – Added meson build to HiGHS (Beale–Orchard-Hays Prize LP/MIP solver), built highspy bindings, integrated into SciPy linprog/milp. Initiated PRIMA integration.
- **OpenBLAS** (contributor) – Ported to meson, added to WrapDB with LAPACK support. Default BLAS/LAPACK for NumPy/SciPy.
- **eOn** (lead maintainer) – Long timescale dynamics: RPC architecture, GP saddle searches, Hessian CI-NEB, ML potentials
- **f2py / NumPy** (maintainer) – Fortran-to-Python interface generator; meson backend, derived type support
- **ASV** (co-maintainer) – Airspeed Velocity: Python benchmarking tool; runner refactor, release management
- **d-SEAMS** (co-creator) – Molecular trajectory analysis engine (JCIM 2020, NumFOCUS funded)
- **metatensor** (contributor) – Atomistic ML data format; DLPack, device-aware execution, Rust core
- **flowy** (contributor) – Probabilistic lava emplacement prediction, 400x speedup (CPC 2025)

TECHNICAL SKILLS

HPC: LAMMPS, GROMACS, MPI, CUDA, Nix, Spack, EasyBuild, Slurm, containers (Docker/Singularity)

Languages: C++, Python, Fortran, Rust, R, Julia, Shell

Tools: CMake, Meson, Git, CI/CD (GitHub Actions, Azure), $\LaTeX$, Emacs

ML/Data: PyTorch, scikit-learn, GPy, Stan, Bayesian optimization, Gaussian processes

PROFESSIONAL MEMBERSHIPS

IOP (MInstP) • RSC (MRSC) • BCS (MBCS) • ACM • APS • IEEE • AIChE